SMART INDIA HACKATHON — SIH26018

BHOOMIVERIFY: INTELLIGENT LAND RECORD DIGITIZATION & VALIDATION SYSTEM

Official Technical Architecture, Functional Compliance & Implementation Report

Status: Prototype Verified • 100% Core & Future Scope Implemented • Date: October 27, 2026

1. EXECUTIVE SUMMARY

Land administration across India faces immense historical friction due to centuries of degraded paper records, handwritten mutations, varying linguistic and department formats, and manual reconciliation bottlenecks. **BhoomiVerify** is an end-to-end autonomous digitization and validation platform engineered to solve **SIH Problem Statement SIH26018**. It ingests degraded paper and digital scans, leverages computer vision (OpenCV) and Indic OCR (Kannada, Hindi, Marathi, English) to extract key land attributes, structures them into relational databases, performs automated multi-point cross-validation against authoritative government land registries, flags duplicates and legal mismatches, and anchors immutable records to a cryptographic ledger.

2. PROBLEM STATEMENT & GROUND CHALLENGES

• **Degraded & Handwritten Scans:** Moisture, tearing, fading ink, and low-contrast paper documents hinder basic OCR.
• **Linguistic Diversity:** Records alternate between English and regional Indic scripts (e.g., Kannada RTC Form 16, Marathi 7/12 Extracts).
• **Fraudulent Duplicates & Encroachment:** Multiple parties registering overlapping claims on the same Survey / Hissa numbers.
• **Discrepancies Between Deeds & Registers:** Sale deeds presenting extents or boundaries differing from authoritative government survey sheets.
• **Laborious Manual Auditing:** Verification backlogs span months per taluk office without digital cross-matching.

3. SIH26018 FUNCTIONAL REQUIREMENTS COMPLIANCE MATRIX

Function / Module	SIH26018 Specification	BhoomiVerify Solution	Status
1. Document Upload	Upload scanned land records, PDFs, or images.	Multi-format uploader (PDF/PNG/JPG) with drag-and-drop & preset pilot samples.	**100% Fully Built**
2. OCR Extraction	Extract names, survey numbers, land area, dates, etc.	OpenCV adaptive preprocessing + Tesseract + Indic OCR (Kannada, Hindi, Marathi).	**100% Fully Built**
3. Record Structuring	Convert unstructured documents into organized records.	Regex & NER extractors parsing owner, relationship, extent, survey/hissa, valuation.	**100% Fully Built**
4. Data Validation	Check required information is present and logically sound.	Multi-rule validation verifying 4-way boundaries, extent metrics, and tax logic.	**100% Fully Built**
5. Duplicate Detection	Identify potentially duplicate land records.	Similarity scoring across survey numbers, hissa, ownership, and taluk/village.	**100% Fully Built**
6. Mismatch Detection	Compare records and flag differences in important fields.	Side-by-side automated engine comparing scanned deeds against reference registry.	**100% Fully Built**
7. Search & Records	Search and retrieve digitized records easily.	Filterable tabular registry with instant search by survey number, owner, status.	**100% Fully Built**
8. Report Generation	Generate reports containing detected errors and audits.	Automated official PDF Certificate generation with QR code verification hash.	**100% Fully Built**

4. ADVANCED & FUTURE-SCOPE CAPABILITIES ALREADY INTEGRATED

BhoomiVerify goes significantly beyond basic OCR by embedding all components listed in the SIH 'Future Scope' section:

Scope Item	Implementation in Prototype	Impact on Land Administration
Multilingual Indic OCR	Native support for Kannada, Hindi, and Marathi along with English revenue forms.	Enables immediate deployment across state revenue offices without re-transcription.
Cadastral GIS Mapping	Integrated Leaflet engine with ESRI Satellite imagery, official district vector boundaries (GeoJSON), and user-drawn query polygons.	Enables spatial verification of ground parcel boundaries against surveyed coordinates.
State Pilot Coverage (4 Districts)	Full administrative hierarchy & reference registers for Davanagere (6 Taluks), Tumakuru (10 Taluks), Shivamogga (7 Taluks), and Mysuru (7 Taluks) — 30 Taluks total.	Demonstrates scalable deployment ready for statewide rollout across all 31 Karnataka districts.
Field Officer Mobile Console	GPS geofencing simulator, real-time spatial proximity verification, and geotagged field photo capture.	Eliminates ghost inspections and empowers on-ground revenue inspectors.
Blockchain Audit Trail	SHA-256 chained cryptographic ledger recording every verification, deed modification, and approval.	Guarantees non-repudiation and prevents unauthorized database tampering.

5. SYSTEM ARCHITECTURE & TECHNICAL SPECIFICATIONS

• **Frontend:** React 18, Vite, Tailwind CSS, Lucide Icons, Leaflet GIS, HTML5 Canvas.
• **Backend:** FastAPI (High performance ASGI asynchronous web server), Python 3.11.
• **Vision & OCR:** OpenCV 4.x (adaptive thresholding, morphology, deskewing), Tesseract 5.x, Indic OCR transliteration engine.
• **NLP & Verification:** Regular Expression Named Entity Recognition, Levenshtein Fuzzy Ratio Discrepancy Matching.
• **Database:** SQLAlchemy ORM, SQLite configured with Write-Ahead Logging (WAL) and 20s lock timeout (drop-in compatible with PostgreSQL / MySQL).
• **Auditing & PDF:** ReportLab PDF Generation Engine with SHA-256 Genesis Block Ledger.

6. VERIFICATION, BENCHMARKING & TEST METRICS

• **Automated Test Suite:** 58 automated unit & integration test cases executed via pytest.
• **Pass Rate: 58 / 58 Passed (100% Pass Rate)**.
• **Test Coverage:** Document OCR, NLP extraction, duplicate detection, mismatch detector, cadastral imports, survey number lookups, Indic Kannada ingestion, and GIS API endpoints.
• **Production Build:** Vite production build transforms 1,546 modules in 22.29s with 0 errors.

7. CONCLUSION & HACKATHON VALUE PROPOSITION

BhoomiVerify fulfills all mandates of Smart India Hackathon problem statement **SIH26018**. By replacing manual paperwork review with automated computer vision, multi-tier discrepancy detection, and cryptographic integrity, the system reduces land record verification time from weeks to seconds. Its complete prototype demonstrates technical feasibility, high user accessibility, and profound social impact for transparent governance.